

CS & IT ENGINEERING

Programming in C

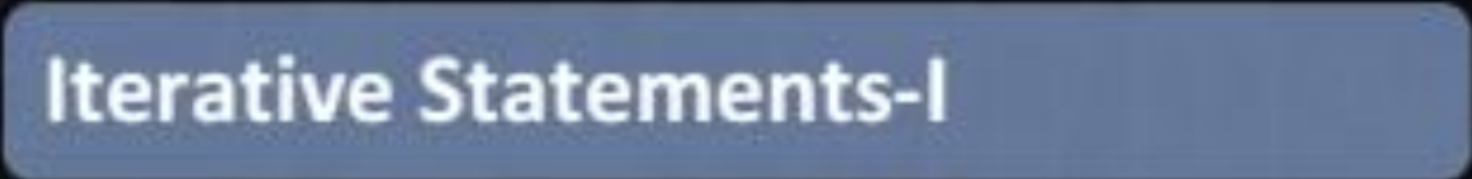
Control Flow Statements
Lec- 02



By- Pankaj Sharma sir



TOPICS TO BE
COVERED



Iterative Statements-I

Right, down

Go right.

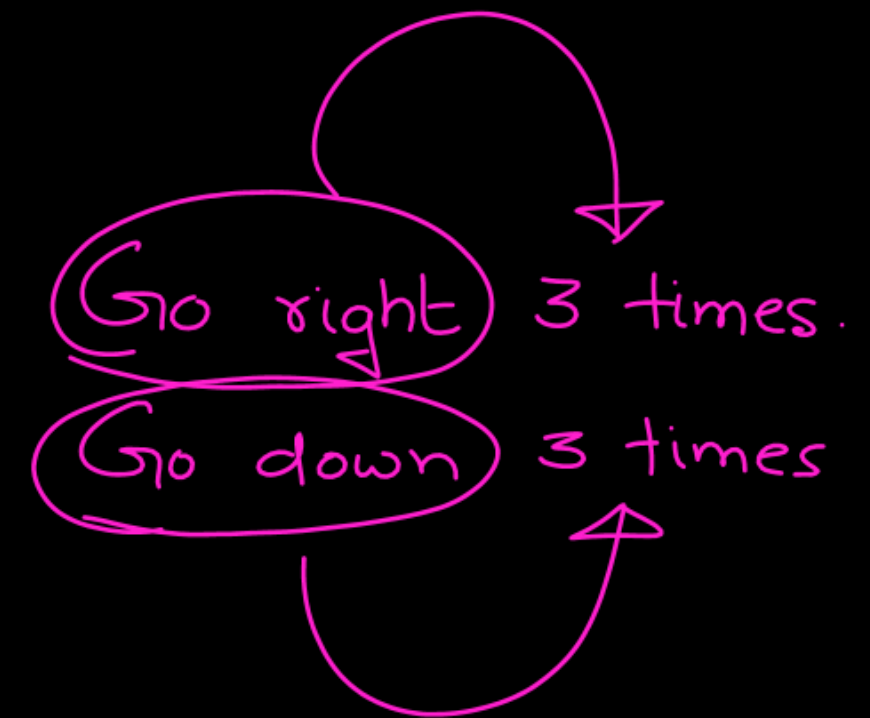
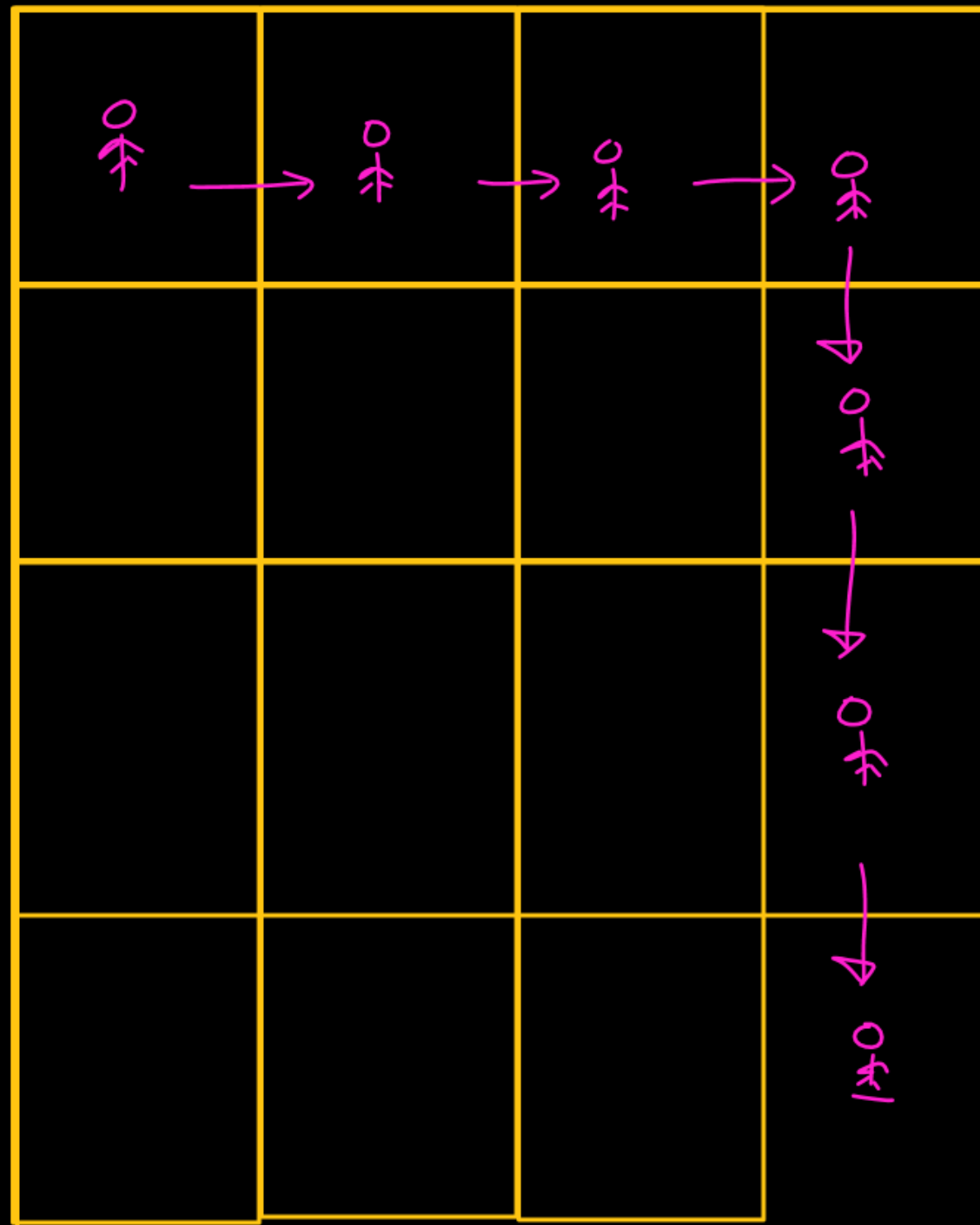
Go right

Go right

Go down

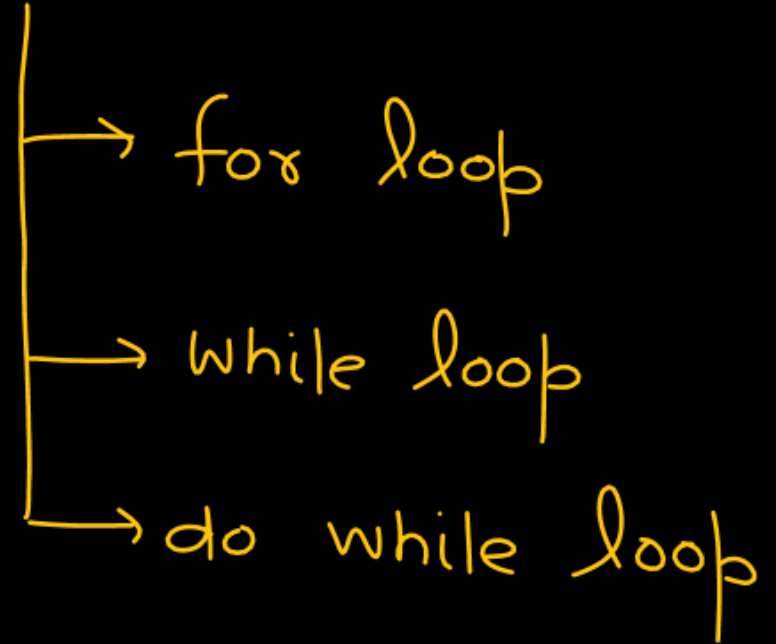
Go down

Go down



Iterative Statement (Repeatition)

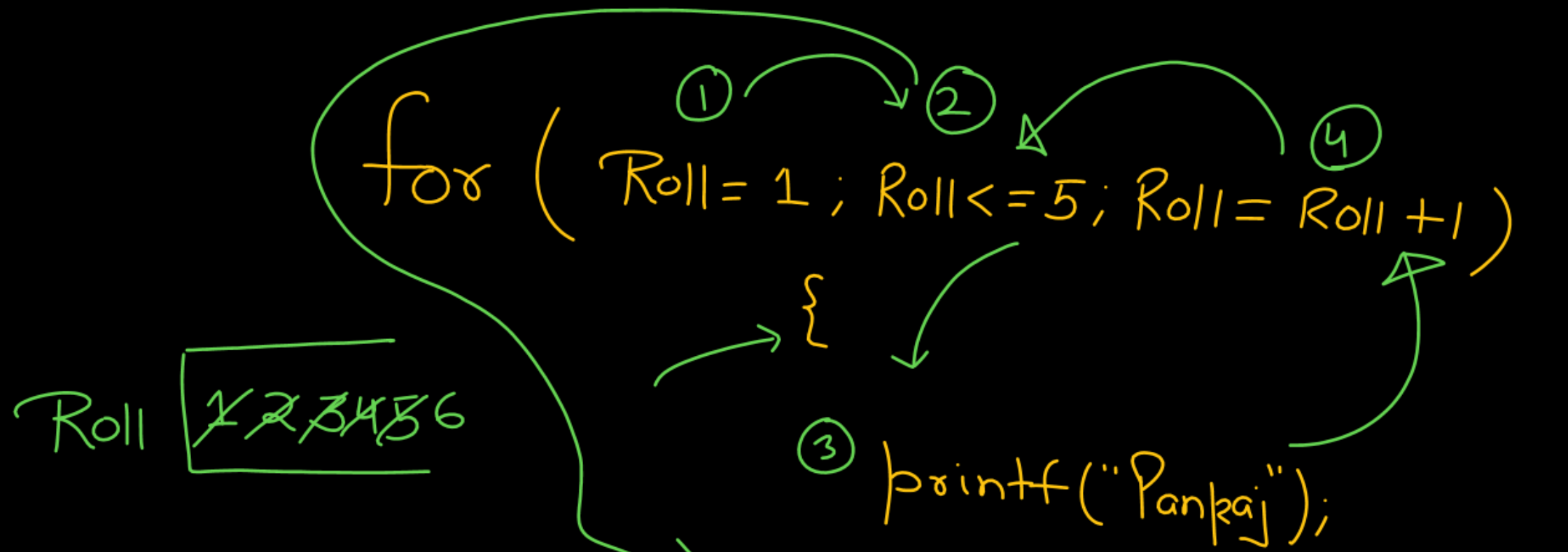
Loops



```

① Roll = 1 ; ② Roll <= 5 ; Roll = Roll + 1
                |
                v
③ printf("Pankaj Sharma");
                ^
                |
                Roll = Roll + 1

```

1 <= 5 → True

2 <= 5 → True

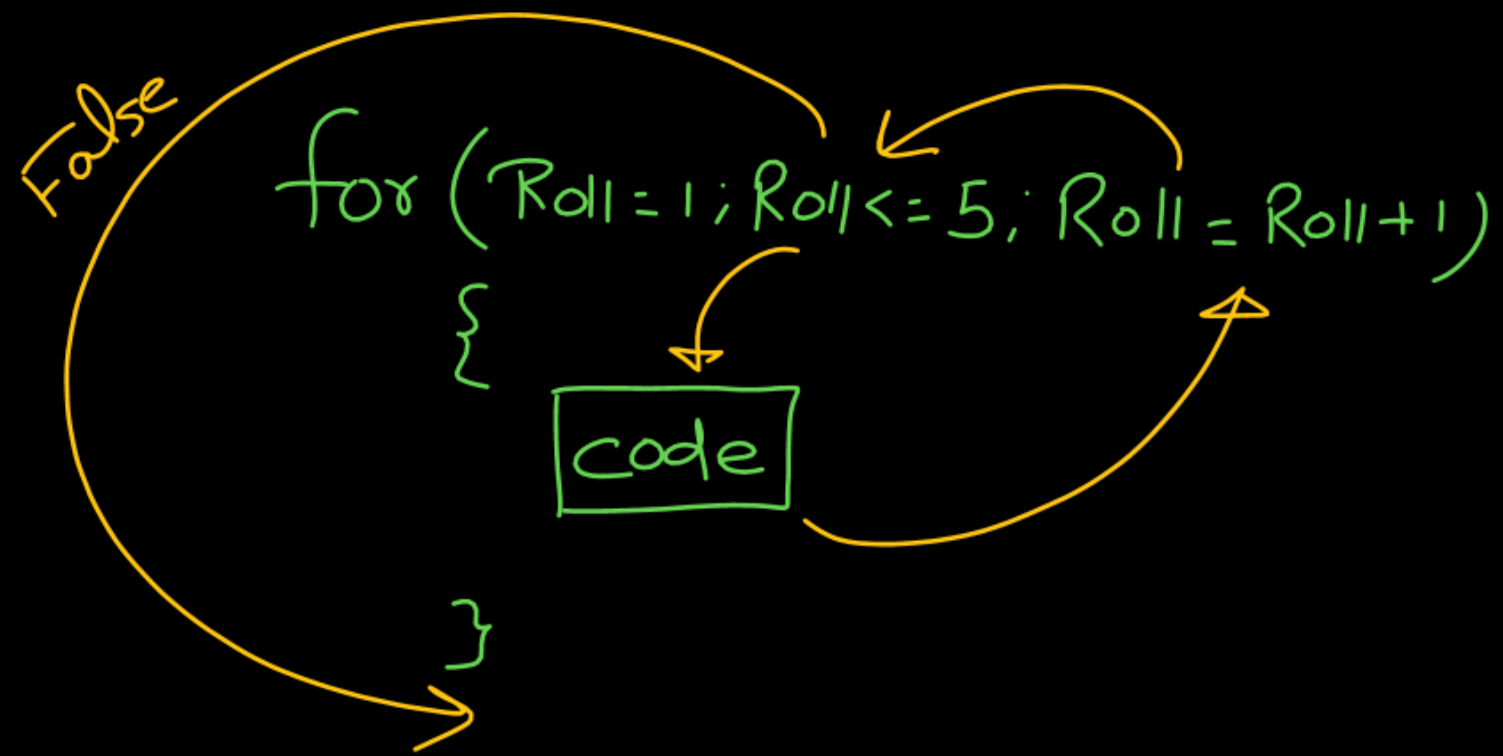
3 <= 5 → True

4 <= 5 → True

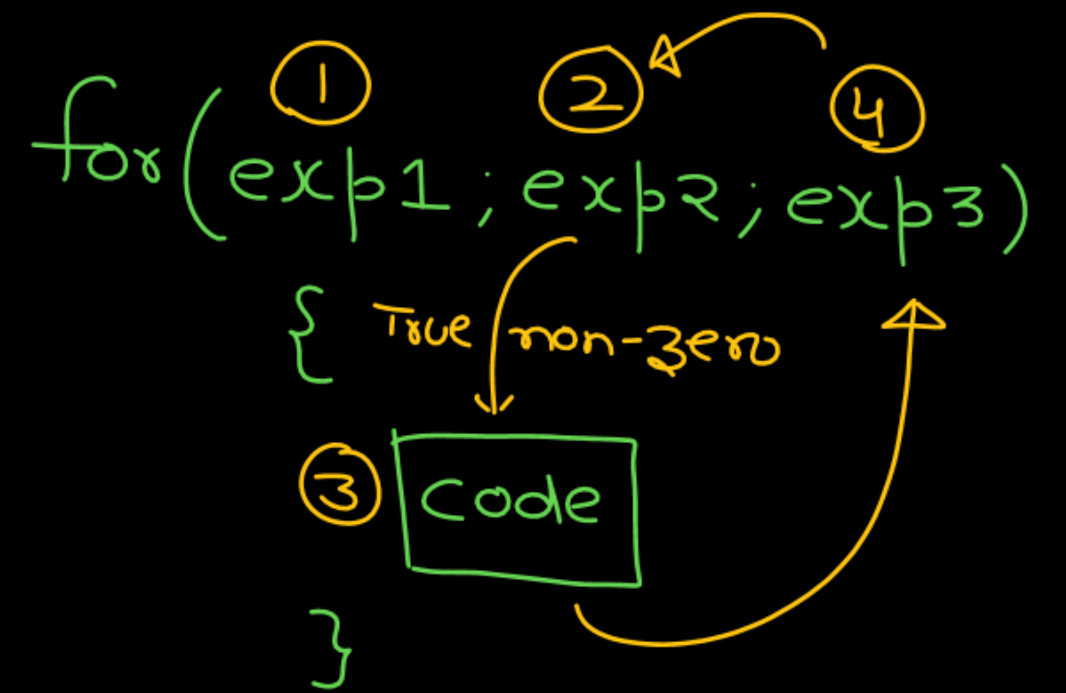
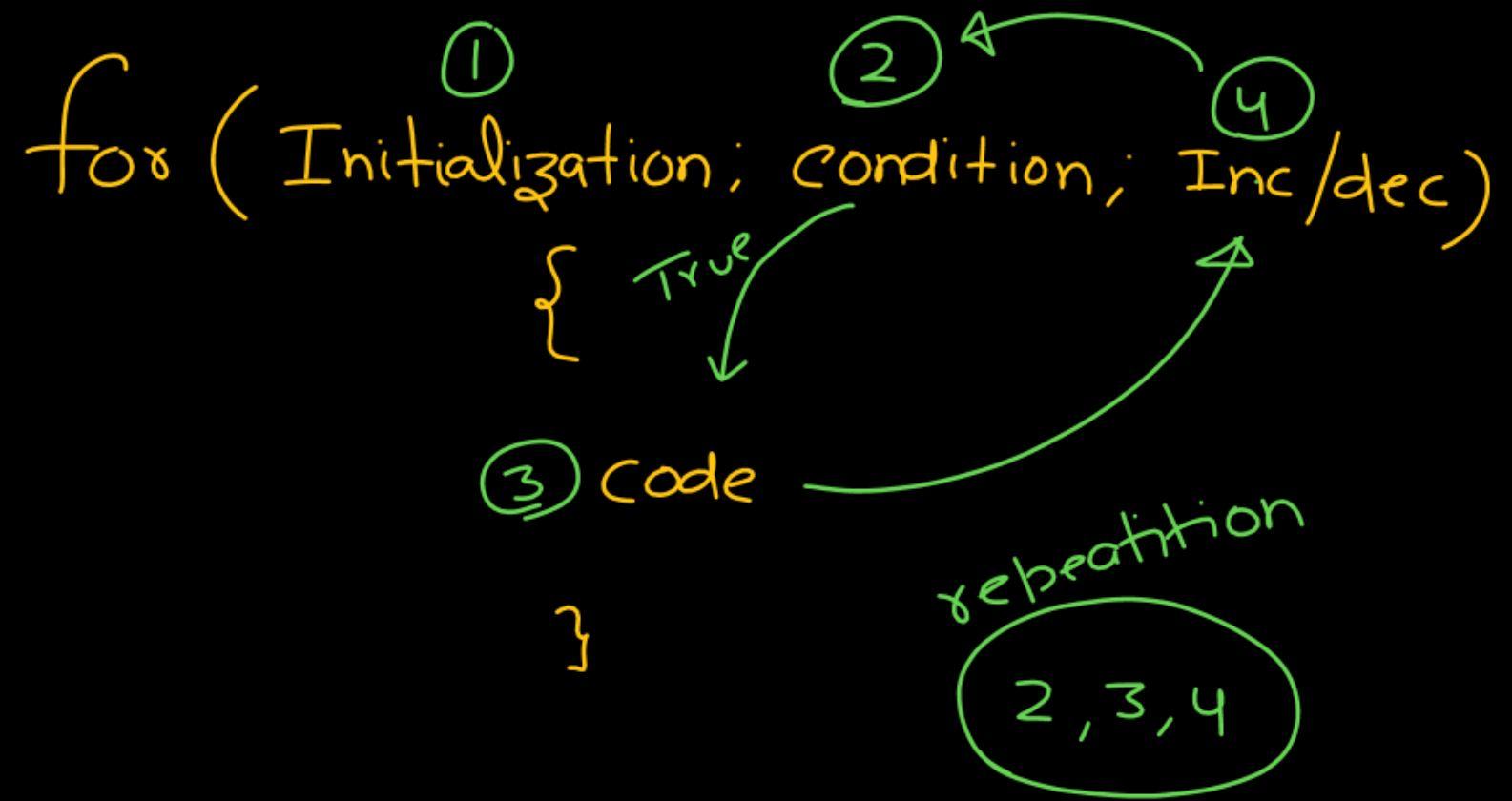
5 <= 5 → True

6 <= 5 → False

Pankaj Pankaj Pankaj Pankaj Pankaj



Roll = 1	1 <= 5 ✓	code
Roll = 2	2 <= 5 ✓	code
Roll = 3	3 <= 5 ✓	code
Roll = 4	4 <= 5 ✓	code
Roll = 5	5 <= 5 ✓	code
Roll = 6	6 <= 5 ✗	



✓

```
for (Roll = 1; Roll <= 5; Roll++)  
{  
    printf("Pankaj");  
}
```

5 times Roll = 1, 2, 3, 4, 5

⇒

```
for (Roll = 6; Roll <= 10; Roll++)  
{  
    printf("Pankaj");  
}
```

⇒

```
for (Roll = -2; Roll <= 2; Roll++)  
{  
    printf("Pankaj");  
}
```

1.

```
for (i=1; k=10; i=i+2)
```

```
printf("/d", i);
```

```
printf("Pankaj");
```

First semicolon



$i \leq 10$

```
for (① i=1; ② k=10; ③ i=i+2)
```

```
{
```

```
④ printf("/d", i);
```

```
}
```

```
printf("Pankaj");
```

$i=1, k=10 \rightarrow \text{True}$ 1

i		
1	$1 \leq 10 \rightarrow \checkmark$	1
3	$3 \leq 10 \rightarrow \checkmark$	3
5	$5 \leq 10 \rightarrow \checkmark$	5
7	$7 \leq 10 \rightarrow \checkmark$	7
9	$9 \leq 10 \rightarrow \checkmark$	9

1 3 5 7 9 Pankaj

int i=1;

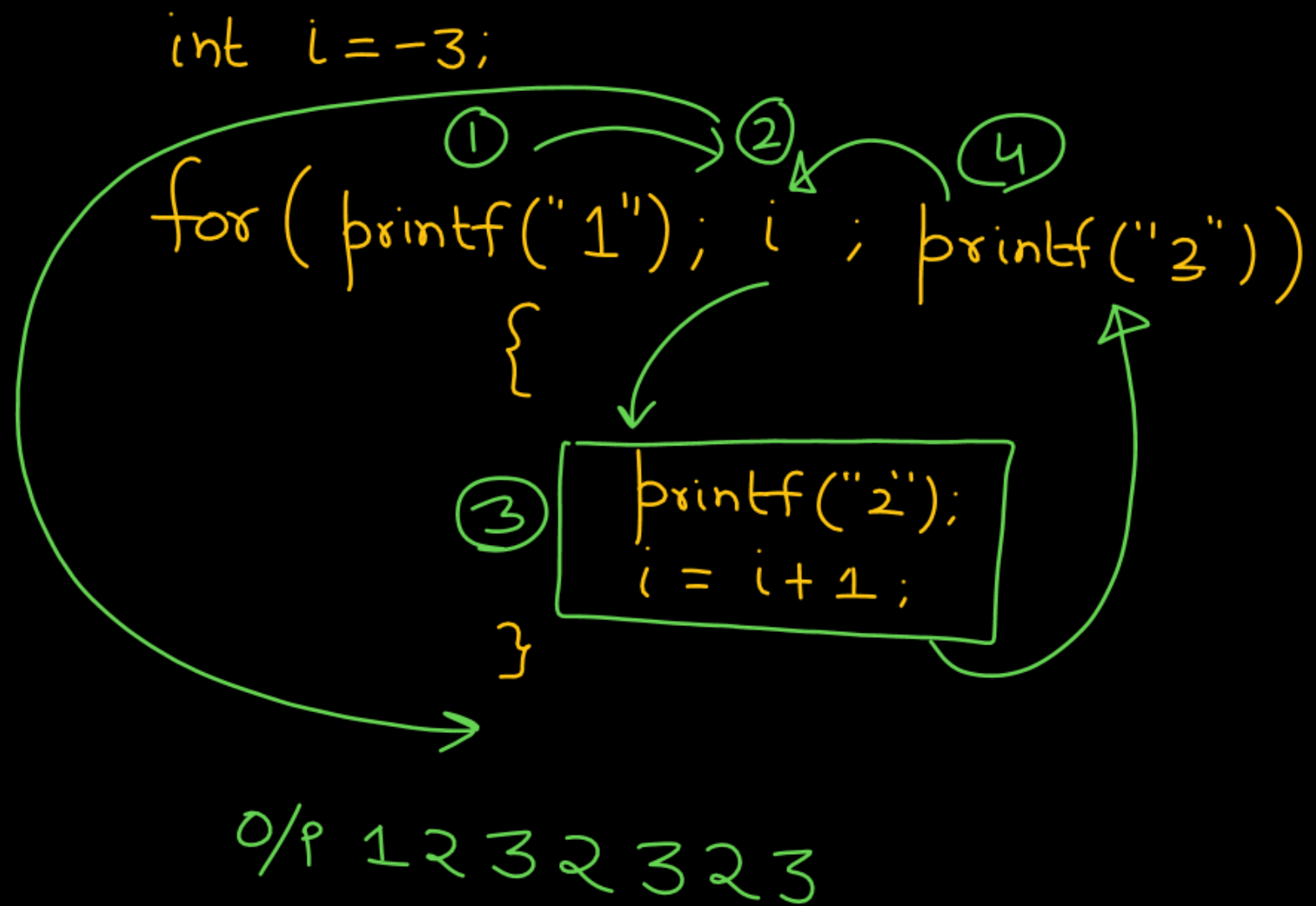
1222222

2.

for(^① printf("1"); ^② i<=5; i++)
{
 printf("2");
}

i	exp2
1	1<=5✓
2	2<=5✓
3	3<=5✓
4	4<=5✓
5	5<=5✓
6	6<=5 → <u>False</u>

3.



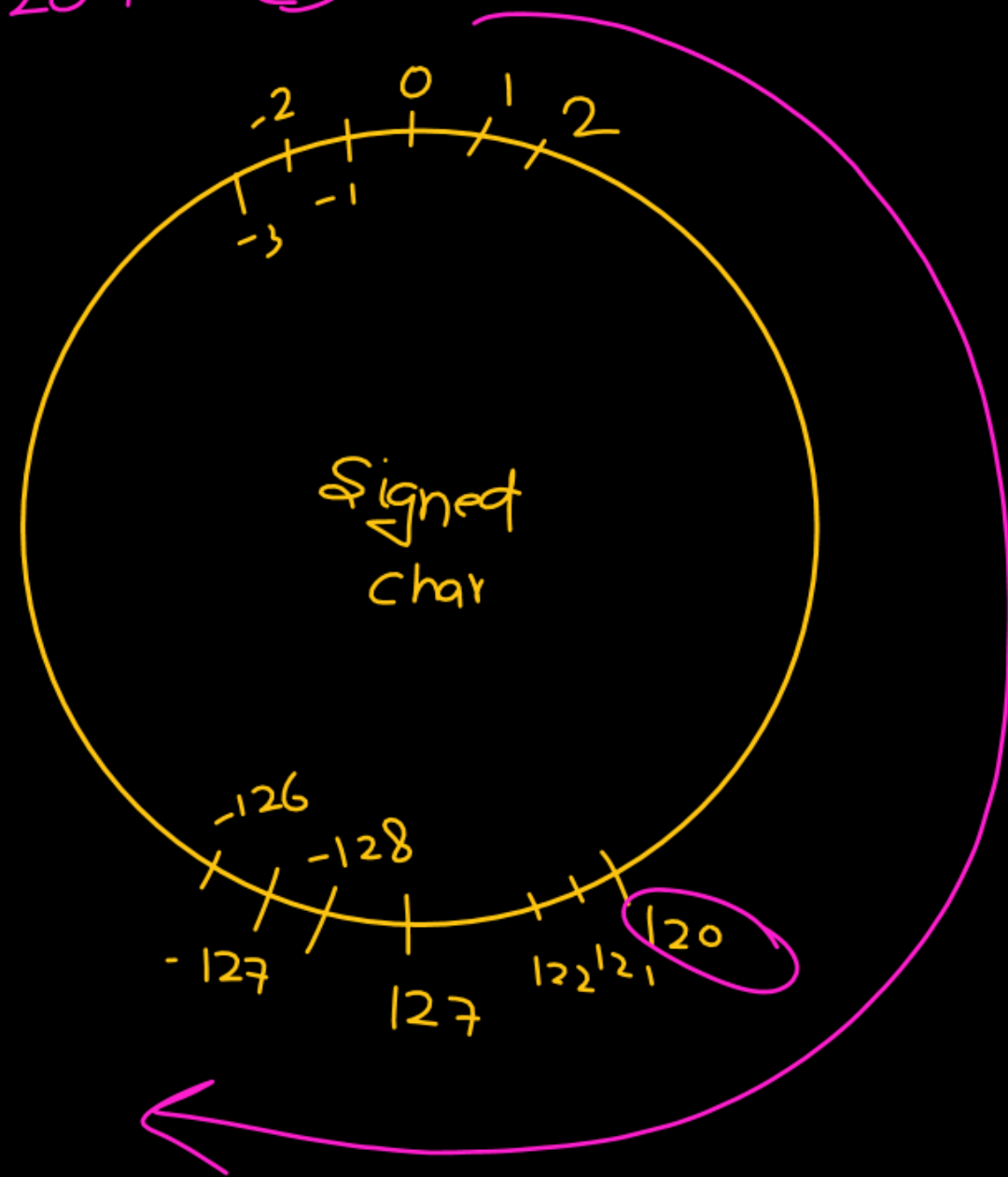
i	
-3	-3 ⇒ True/non-zero ✓
-2	-2 ⇒ True ✓
-1	-1 ⇒ True ✓
0	0 ⇒ False

4. char ch;
 Signed
 for(^①ch=120; ^②ch; ch=ch+1)
 {
 printf("Pankaj");
 }

ch	
120	120 ✓
121	
122	
⋮	
127	
-128	
-127	
⋮	
-1	
0	X

8 + 128 = 136

$$127 - 120 + 1 = 8$$



2.

```
char ch;
```

```
for (ch = 11; ch; ch = ch + 2)
```

$$\{$$

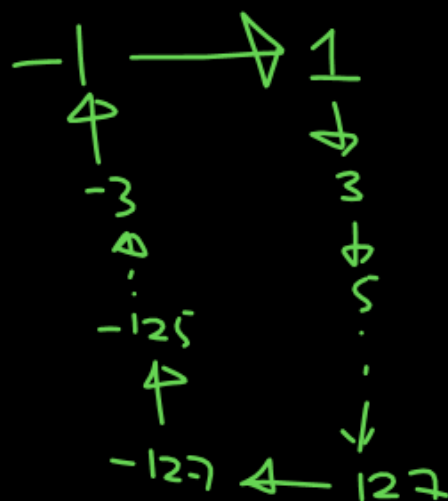
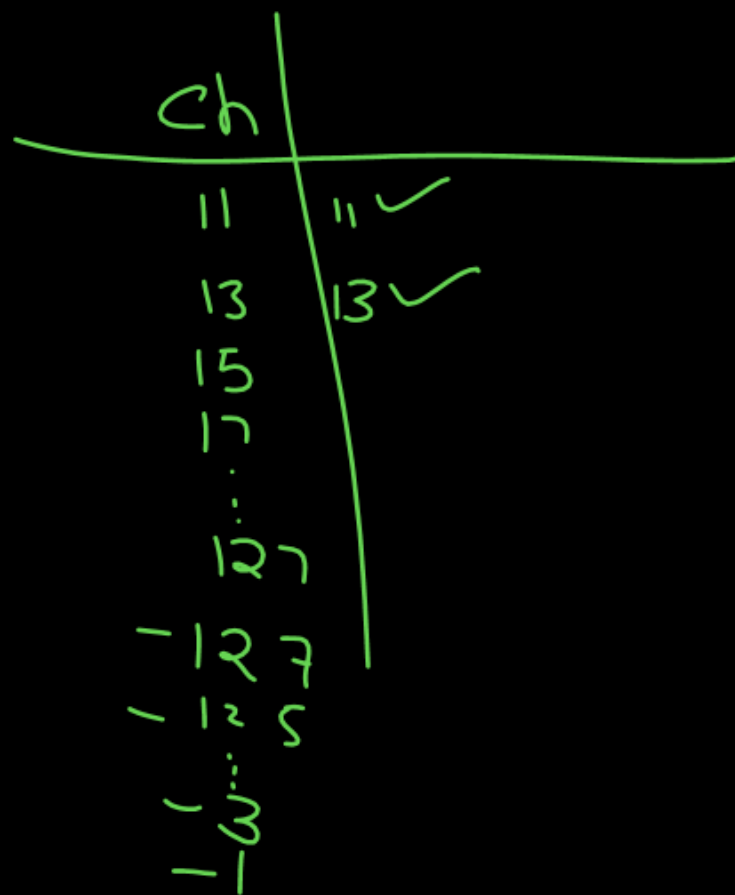
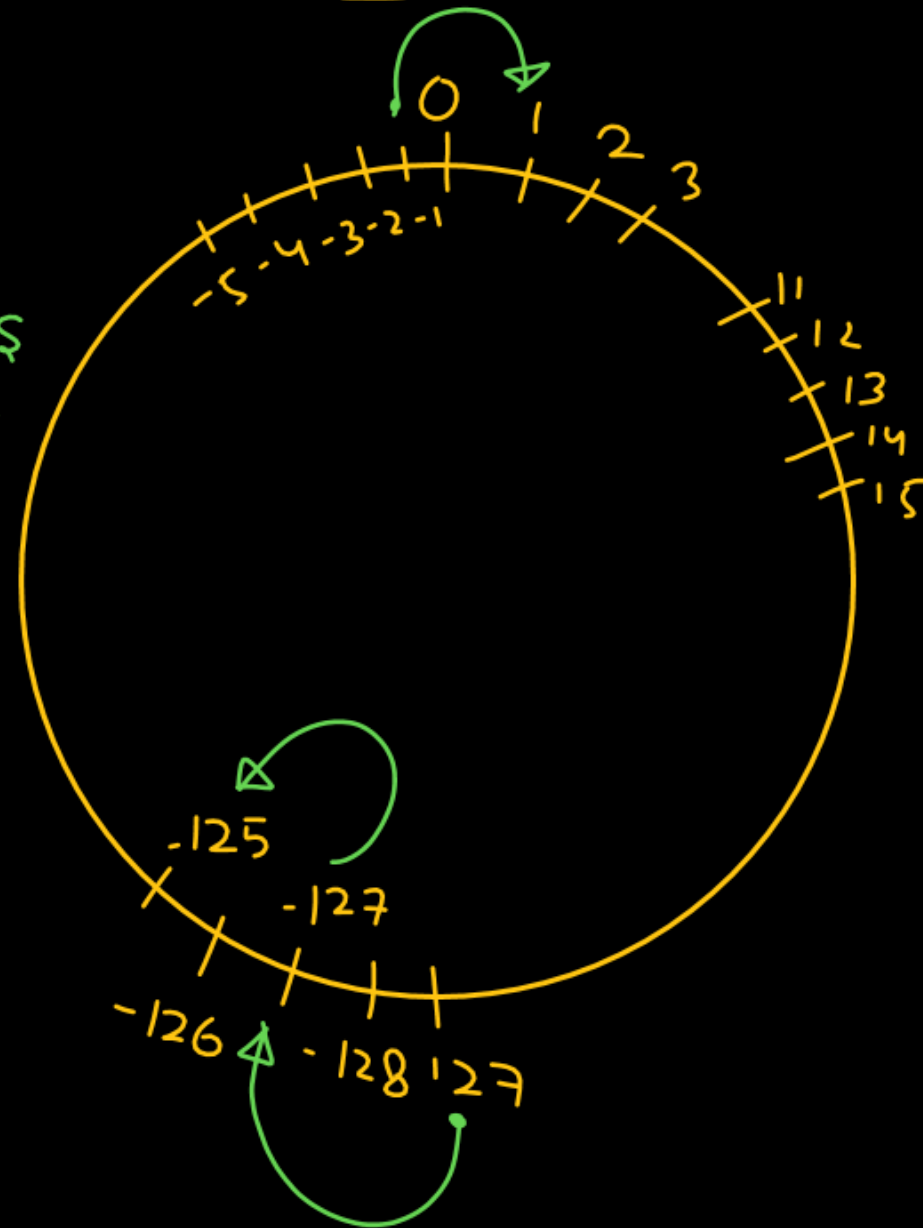
```
printf("Pankaj");
```

3

∞ times

Sunday

[5:00 - 7:00 PM]
[8:30 - 10:30 PM]



3.

```
void main()  
{
```

```
char ch;
```

```
for (ch = 1; ch; ch = ch + 2) (;
```

First semi
colon

⇒

```
for (ch = 1; ch; ch = ch + 2)
```

```
{
```

```
;
```

```
}
```

```
printf("Pankaj");
```

```
}
```

4.

```
char i = -1;  
for (i++; i++; i++)  
{  
    printf("Pankaj");  
}  
0 times
```

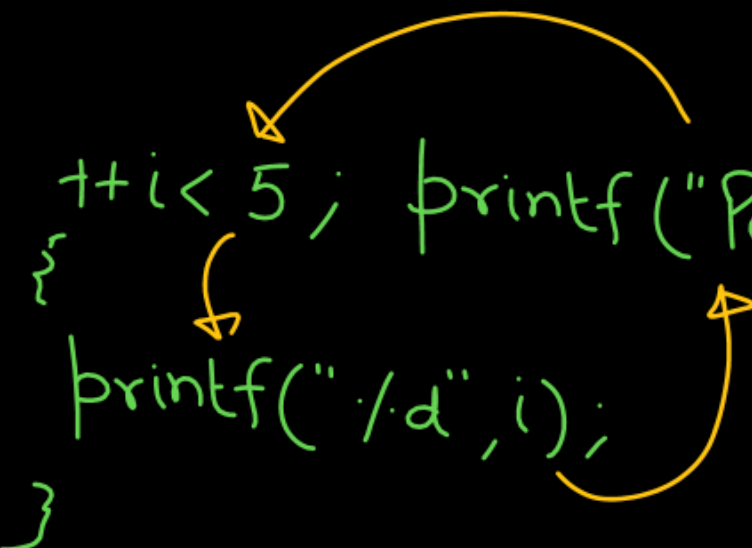
Diagram: A green arrow points from the first 'i++' to the second 'i++'. Above the arrow is a '-1' and below it is a '0'. To the right of the arrow is the text 'eval. => 0'.

Post-inc => use
then inc.

5.

```
void main() {  
    int i;
```

```
    for (i = 1; ++i < 5; printf("Pankaj"))  
    {  
        printf("%d", i);  
    }  
}
```



i
1
2
3
4
5

2 < 5 → True

3 < 5 → True

4 < 5 → True

5 < 5 → False

O/P: 2 Pankaj 3 Pankaj 4 Pankaj

Q int i

for(i=1; i++<5; printf("Pankaj"))
{
printf("%d", i);
}

a) i < 5
b) i = i + 1

i [1 2 3 4 5]

1) 1 < 5 $\Rightarrow$ True

2 < 5 $\Rightarrow$ True

3 < 5 $\Rightarrow$ True

4 < 5 $\Rightarrow$ True

5 < 5 $\Rightarrow$ False

2 Pankaj 3 Pankaj 4 Pankaj 5 Pankaj

Q1

```
for (2; 3; 4)
{
    printf("Ranjay");
}
```

∞

Q2

```
for (2; 0; 3)
{
    printf("Hello");
}
```

0 times

Q3

```
for (1; -1; 3)
{
    printf("Hello");
}
```

∞

non-zero/True

```
for (exp1; exp2; exp3)
{
    code
}
```

~ All three expressions are optional


```
int i;  
for(i=1; i<=5; i++)  
{  
    printf("%d", i);  
}
```

```
int i;  
i=1;  
for(  
    ; i<=5; i++)  
{  
    printf("%d", i);  
}
```

→ Valid


```
for(i=1; i<=5; i++)  
{  
    printf("/d", i);  
}
```

$\Rightarrow$

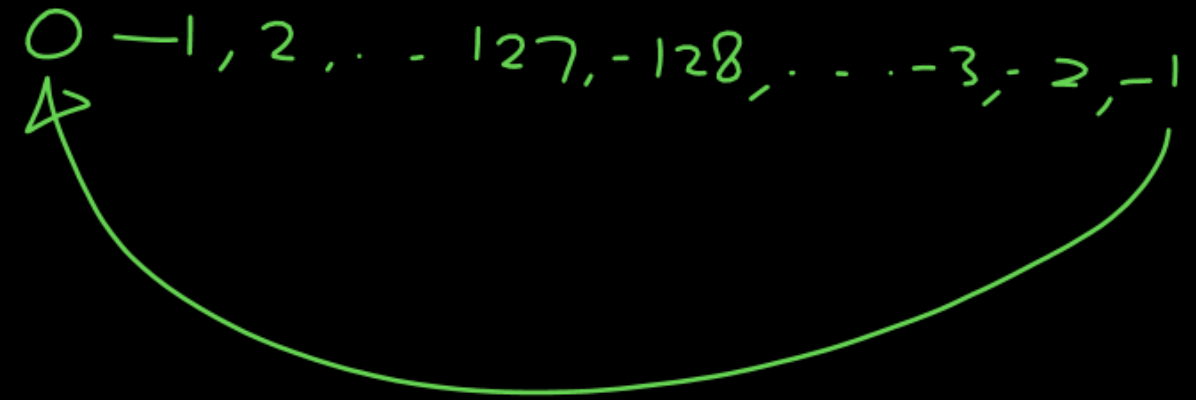
```
for(i=1; i<=5; )  
{  
    printf("/d", i);  
    i++;  
}
```

```
for(i=1; i<=5; i++)  
    printf("/d", i);
```

⇒

```
i=1;  
for(    ①  
    ; i<=5;    )  
{  
    printf("/d", i);  
    i++;  
}
```

```
char i;  
for (i=0; ; i++)  
{  
    printf("%d", i);  
}
```



∞ loop

```
for ( ; ; )
```

```
{
```

```
printf("%d", i);
```

```
}
```

1.

```
for (i=1; i<=3; i++)  
{  
    code  
}
```

$i = 1, 2, 3$
code
3 times

2.

```
for (i=1; i<=10; i++)  
{  
    code  
}
```

10
 $i =$ 1
2
⋮
10
11

$1 <= 10$
 $2 <= 10$
 $10 <= 10$
 $11 <= 10$

$i = 1, 2, 3, \dots, 10$
 $1 \rightarrow 10$
 $\Rightarrow 10 - \cancel{1} + \cancel{1}$
10 times

3. for (i=1; i<=n; i++)
 {
 printf("Pankaj");
 }

n times

4. for (i=1; i<=n; i=i+2)
 {
 printf("Pankaj");
 }

$\left\lceil \frac{n}{2} \right\rceil$ times

Exact answer :

n=5

i=1, 3, 5

③ times

n=6

i=1, 3, 5

③ times

n=7

i=1, 3, 5, 7

④ times

$$\left\lceil \frac{7}{2} \right\rceil = \left\lceil 3.5 \right\rceil = ④$$

$$\left\lceil \frac{n}{2} \right\rceil$$

S:

```
for (i=1; i<=n; i=i*2)
{
    printf("Pankaj");
}
```

① $n = 100$
 $i = 1, 2, 4, 8, 16, 32, 64, 128$
7 times

② $n = 64$
 $i = 1, 2, 4, 8, 16, 32, 64$
last value of i
7 times

$i = 1, 2^1, 2^2, 2^3, \dots, 2^k$
(k+1) times

$(\lfloor \log_2 n \rfloor + 1)$ times

last value of $i \leq n$

$2^k \leq n$
 $\log(2^k) \leq \log n$
 $k \log 2 \leq \log n$
 $k \leq \log_2 n$

$$\lceil \log_2 7 \rceil \Rightarrow \lceil 2.8 \rceil = 3$$

$$\lfloor \log_2 7 \rfloor \Rightarrow \lfloor 2.8 \rfloor = 2$$

```
for (i = 1; i <= n; i = i * 3)
{
    printf("Pankaj");
}
```

$$\left(\lfloor \log_3 n \rfloor + 1 \right)$$

